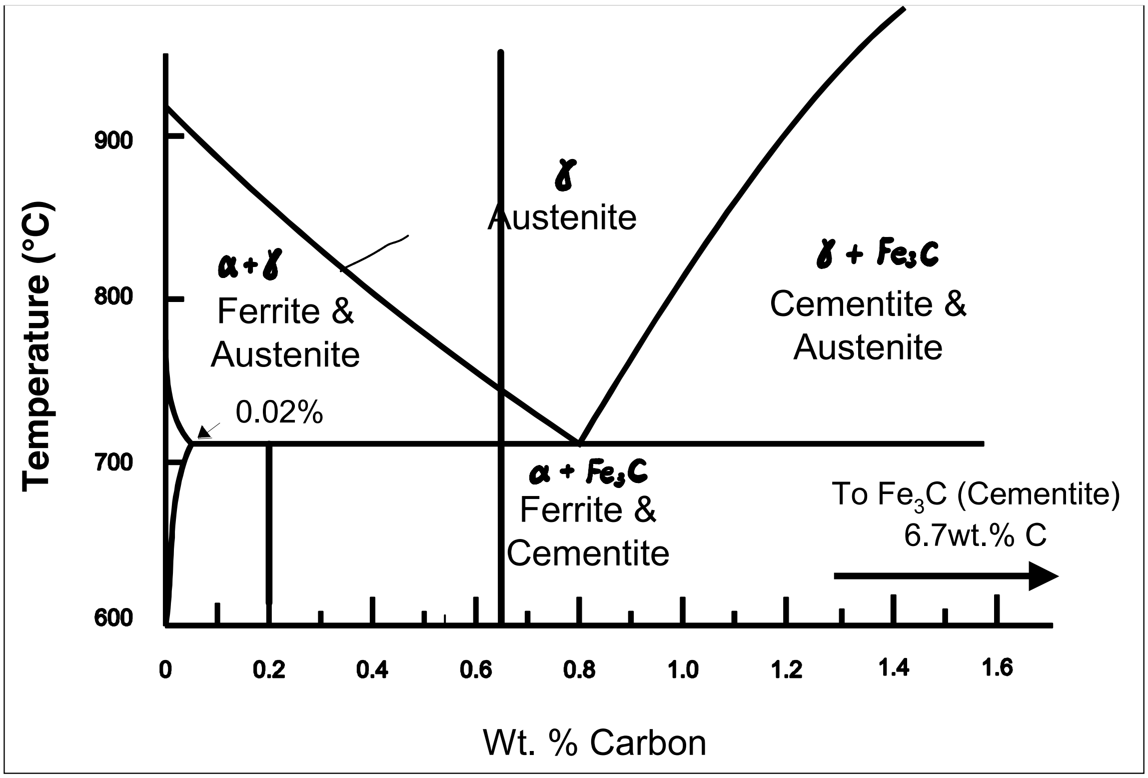
AERO50005 TUTORIAL 2

The key aerospace alloys

Questions

Question 1. Referring to the Fe-Fe_3C phase diagram below, answer the following questions:



Consider 2.5kg of austenite containing 0.65 wt.% C, cooled to below 723°C

(a) What is the pro-eutectoid phase (phase formed before eutectoid reaction);

ferrite ~~+ austenite~~

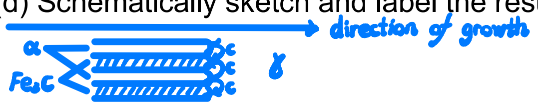
(b) How many kilograms each of total ferrite and cementite form (at just below 723°C);

$f_\alpha + f_{Fe_3C} = 1$, $f_\alpha\ 0.02wt\% + f_{Fe_3C}\ 6.7wt\% = 0.65wt\%$ $\therefore f_{Fe_3C} = 9.43\%$, $f_\alpha = 90.57\%$ $\therefore m_{Fe_3C} = 0.236kg$, $m_\alpha = 2.264kg$

(c) How many kilograms each of pearlite and pro-eutectoid phase form (at just below 723°C);

$f_\alpha + f_{\alpha+Fe_3C} = 1$, $f_\alpha\ 0.02wt\% + f_{\alpha+Fe_3C}\ 0.8wt\% = 0.65wt\%$ $\therefore f_{\alpha+Fe_3C} = 80.77\%$, $f_\alpha = 19.23\%$ $\therefore m_{\alpha+Fe_3C} = 2.019kg$, $m_\alpha = 0.481kg$

(d) Schematically sketch and label the resulting microstructure.



Question 2.

You have been given a steel of unknown Carbon content. The steel has been normalised (heated into the Austenite phase field and slowly cooled) and it was found that the microstructure consisted of pro-eutectoid ferrite and pearlite; the mass fractions

α α + Fe3C

of these two micro constituents are 0.286 and 0.714 respectively. Determine the concentration of Carbon in this alloy.

f_α 0.02 wt% + f_{Fe_3C} 0.8 wt% = 0.577 wt%

Question 3.

Often the properties of multiphase alloys may be approximated by a law of mixtures relationship. i.e: $P_{(Alloy)} = P_{M1} \cdot f_{M1} + P_{M1} \cdot f_{M1}$

$P_{(Alloy)}$ = Overall Property of the Alloy
P_{M1} = Property of Micro constituent 1
P_{M2} = Property of Micro constituent 2
f_{M1} = Fraction of Micro constituent 1
f_{M2} = Fraction of Micro constituent 2

$f_\alpha + f_{\alpha+Fe_3C} = 1$

f_α 0.02 wt% + $f_{\alpha+Fe_3C}$ 0.8 wt% = 0.2%

$\therefore f_{\alpha+Fe_3C} = \frac{3}{13}$, $f_\alpha = \frac{10}{13}$

Employ this relationship to determine the approximate tensile strength of α 0.2 wt.% C steel. Assume that the tensile strength of pro-eutectoid ferrite (α) is 400 and pearlite ($\alpha+Fe_3C$) is 850.

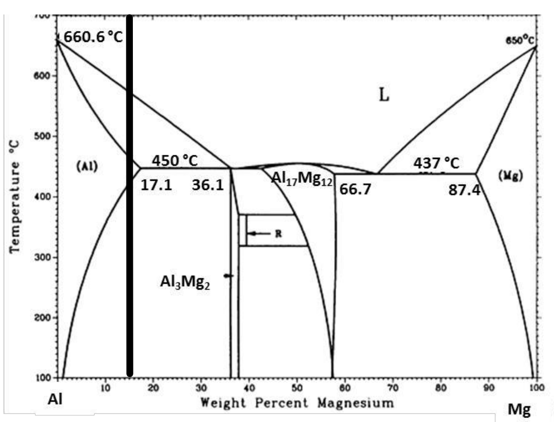
$P = P_\alpha f_\alpha + P_{\alpha+Fe_3C} f_{\alpha+Fe_3C} = 503.85$

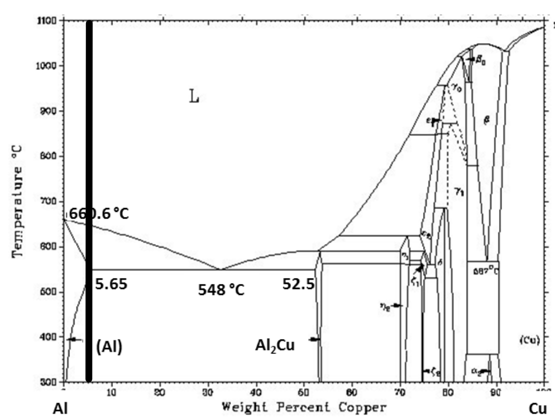
Question 4.

Most aerospace Al-alloys are precipitation strengthened. One factor determining the efficiency of precipitation strengthening is the fraction of precipitates. With reference to the phase diagrams below, answer the following questions.

(4a) Calculate the fraction of precipitate phase at 100°C in binary Al-5wt%Cu and Al-15wt%Mg

(4b) Suggest a suitable solution heat treatment temperature in each case.





(4a) $f_{Al} + f_{Al_3Mg_2} = 1$. f_{Al} 1 wt% + $f_{Al_3Mg_2}$ 36.1 wt% = 5 wt% $\therefore f_{Al} = 88.6\%$ $\therefore f_{Al_3Mg_2} = 11.4\%$

$f_{Al_2Cu} = 1$. f_{Al_2Cu} 52.5 wt% = 5 wt% $\therefore f_{Al_2Cu} = 9.52\%$

(4b) 450°C and 548°C   400°C < T < 470°C and 500°C < T < 580°C

Question 5.

List factors other than precipitate phase fraction that have a significant influence on the efficiency of precipitation strengthening. Explain how strength is influenced by each factor.

- precitation size and spacing
- precitation location
- mechanical properties of the precipitation phase
- matrix precipitate interface (e.g. coherent)